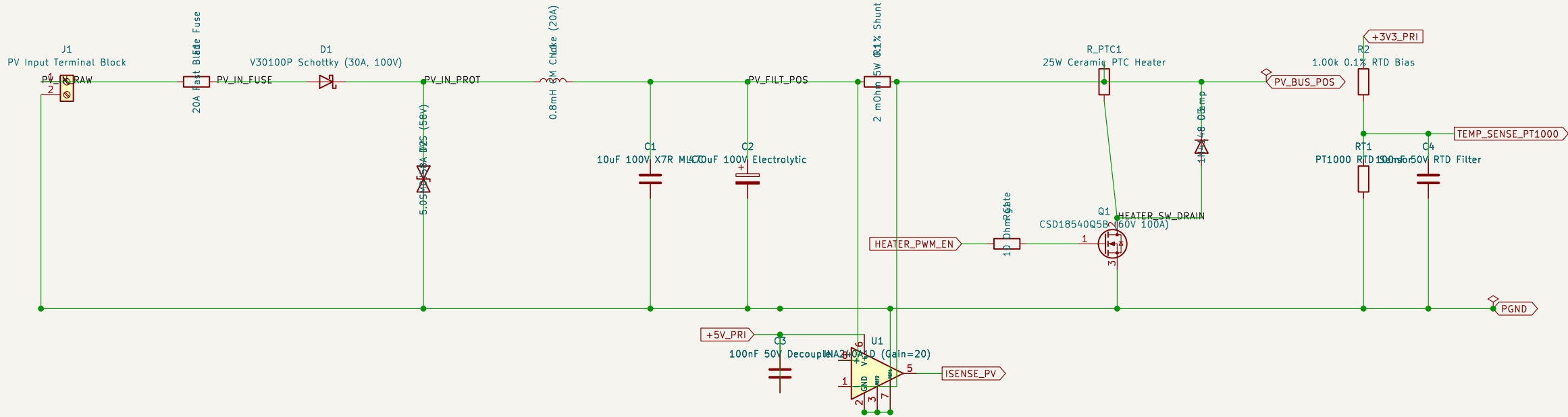


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PV INPUT & HIGH-ALTITUDE COLD-START SUBSYSTEM:
- Input Specification: 40V – 60V DC Nominal, 20A Max Continuous Current
- Transient Immunity: 5.0SMDJ58A 5kW Bidirectional TVS (IEC 61000-4-5 Level 4)
- Sub-Zero Survival: 25W High-Power PTC Ceramic Heater (Autonomous -40C Cold-Start Pre-Warm)
- Precision Telemetry: TI INA240A1D Bidirectional Zero-Drift Shunt Amplifier (CMTI > 100 kV/us)



Operating Envelope: -40C to +50C, 5,000m HAA Ladakh (54.05 kPa)

DRDO SIH26049 | HAR-SMI-500

Sheet: /PV Input & Protection/
File: 01_PV_Input_PTC.kicad_sch

Title: PV Input Stage, Protection, PTC Heater & Current Sense

Size: A3 Date: 2026-09-04

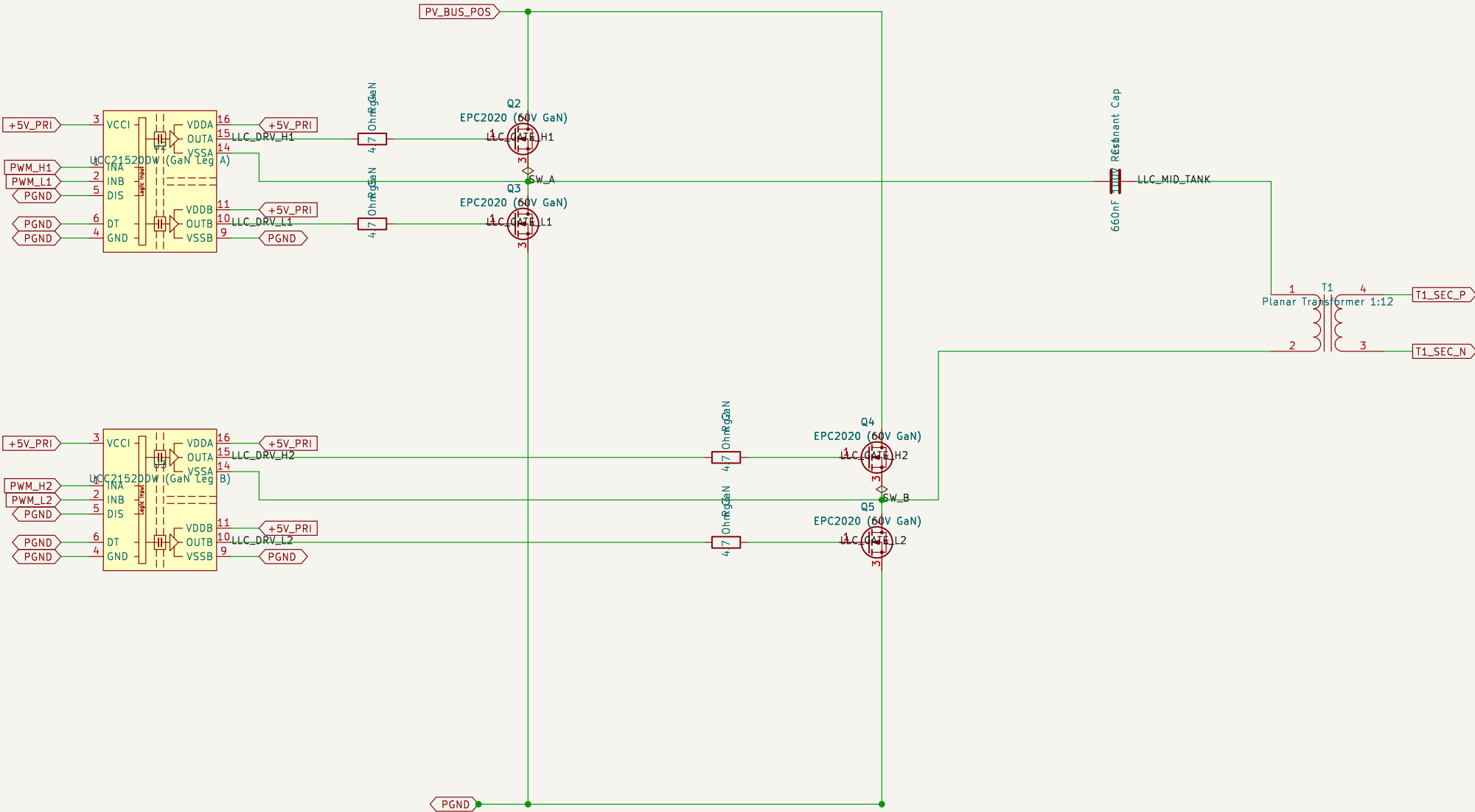
Rev: 3.0-HAA

KiCad E.D.A. 9.0.6

Id: 2/7

PRIMARY POWER CONVERSION: GaN FULL-BRIDGE LLC RESONANT STAGE

- Resonant Frequency: $f_{res} = 100\text{ kHz} - 250\text{ kHz}$ (ZVS soft-switching across full load range)
- Switches: 4x EPC2020 GaN HEMTs (60V, 90A, $R_{on} = 2.2\text{ m}\Omega$, $Q_g = 14\text{ nC}$)
- Isolated Drivers: 2x TI UCC21520DW (5.7 kVrms galvanic isolation barrier, 4A/6A source/sink)
- Planar Transformer T1: Ratio 1:12, Interleaved windings, Primary current = 14.8A RMS



Operating Envelope: -40°C to $+50^{\circ}\text{C}$, 5,000m HAA Ladakh (54.05 kPa)

DRDO SIH26049 | HAR-SMI-500

Sheet: /LLC Converter & GaN/
File: 02_LLC_Primary_GaN.kicad_sch

Title: GaN Full-Bridge ZVS LLC Converter & Planar Transformer

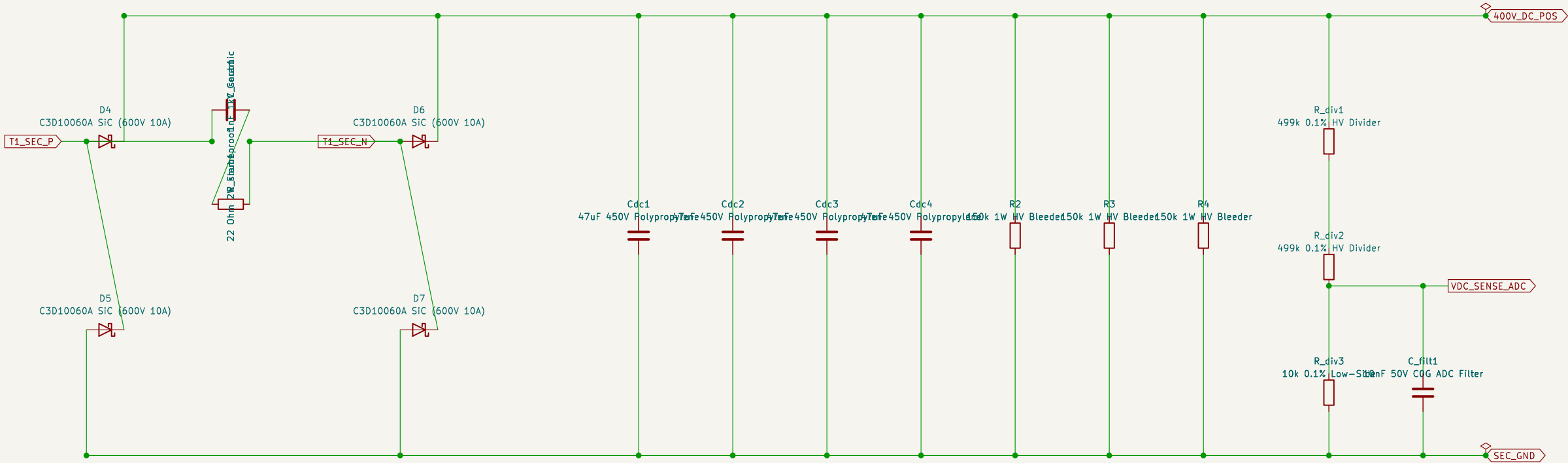
Size: A3
KiCad E.D.A. 9.0.6

Date: 2026-09-04

Rev: 3.0-HAA

Id: 3/7

HIGH-VOLTAGE SECONDARY RECTIFICATION & DC-LINK FILTERING:
- Rectifier Topology: Full-Bridge Fast-Recovery (4x Cree/Wolfspeed C3D10060A 600V SiC Schottky)
- DC-Link Capacitance: 4x 47uF 450V High-Reliability Metalized Polypropylene Film (Low ESR < 8 mOhm)
- Safety Bleeders: Redundant 1W thick-film resistors discharging 400V DC bus in < 30 seconds
- Precision Sense Divider: 100:1 matched low-drift string feeding 12-bit ADC with OVP clamping



Operating Envelope: -40C to +50C, 5,000m HAA Ladakh (54.05 kPa)

DRDO SIH26049 | HAR-SMI-500

Sheet: /Rectifier & DC-Link/
File: 03_Secondary_Rectifier_DCLink.kicad_sch

Title: High-Voltage Secondary Rectifier & 400V DC-Link

Size: A3 Date: 2026-09-04

Rev: 3.0-HAA

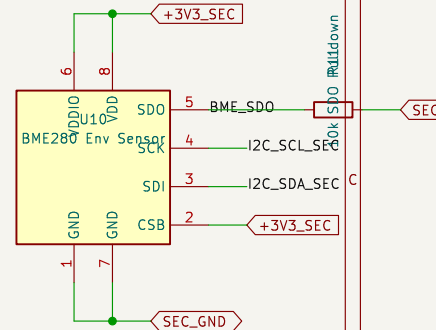
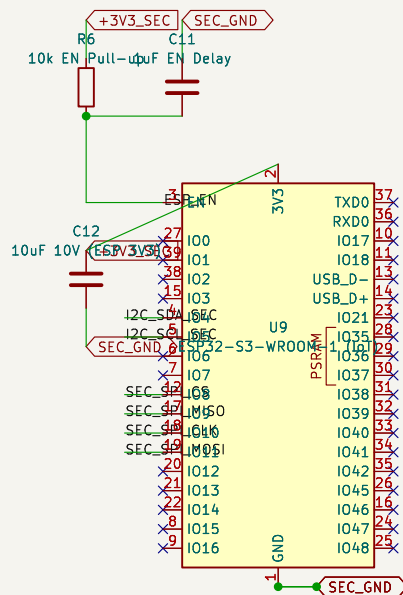
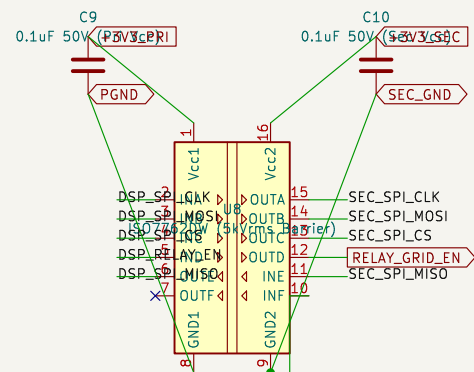
KiCad E.D.A. 9.0.6

Id: 4/7

- Full-Bridge Inverter: 4x Wolfspeed C3M0075120K 1200V SiC MOSFETs (TO-247-4 Kelvin source package)
- Isolated Drivers: 2x TI UCC21520DW with dedicated +15V/-4V drive rails for high dV/dt immunity (>100 V/ns)
- High-Order LCL Filter: $L_{\text{inv}} = 1.8 \text{ mH}$, $C_{\text{filt}} = 4.7 \text{ uF}$ (305V AC), $L_{\text{grid}} = 0.6 \text{ mH}$, $R_{\text{damp}} = 3.3 \text{ Ohm } 5\text{W}$
- Grid Safety Disconnect: Redundant Omron G2RL-1A relays disconnect in <20 ms upon loss of mains

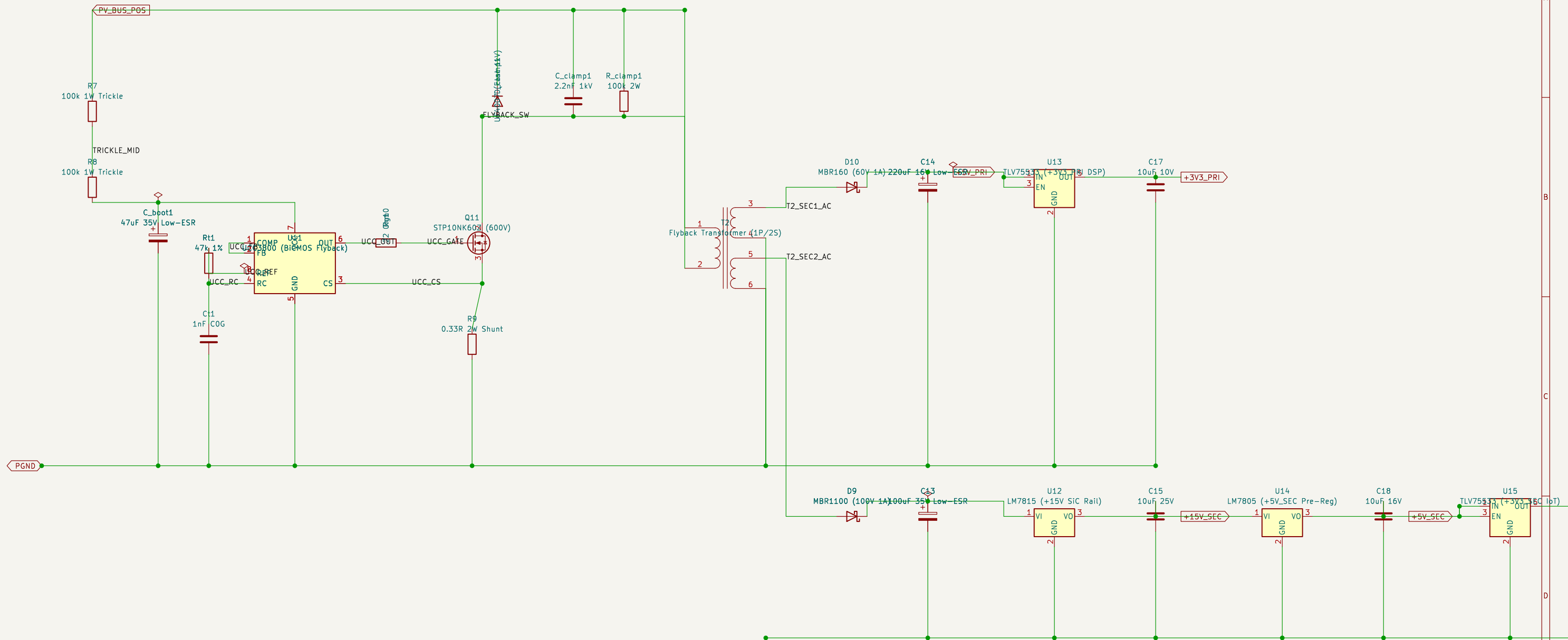


- Primary Domain: Texas Instruments TMS320F280049C Piccolo DSP (100MHz C28x, 100ps HRPWM)
- Secondary/IoT Domain: Espressif ESP32-S3-ROOM-1 (Dual Xtensa LX7 240MHz, BLE 5.0 + Wi-Fi 4)
- Galvanic Isolation Barrier: TI ISO7762DW (6-Channel, 5 kVrms reinforced, CMTI > 100 kV/us)
- Environmental Telemetry: Bosch Sensortec BME280 (High-precision barometric pressure & temp)



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- Controller: TI UCC3800 Current-Mode PWM (Low startup current <100 μ A, 12V UVLO turn-on)
- Primary Flyback Switch: STMicroelectronics STP10NK60Z (600V, $R_{on} = 0.65 \Omega$, Zener-protected gate)
- Transformer T2: 1-Primary / 2-Secondary Toroid (Reinforced triple-insulated secondary windings)
- Regulated Voltage Rails: +15V SiC driver, +5V/+3V3 Primary MCU, +5V/+3V3 Isolated IoT domain



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